

CFET Beyond 3 nm: SRAM Reliability under Design-Time and Run-Time Variability

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ABSTRACT This work investigates the reliability of complementary field-effect transistors (CFETs) by addressing both design-time variability arising from process variations and run-time variability due to temperature and aging effects. A rigorously calibrated TCAD model, validated against experimental CFET data, is employed to quantify the impact of metal gate granularity (MGG)-induced work function variation (WFV) and random dopant fluctuation (RDF) on key device parameters, including the threshold voltage (V_{TH}), on-state current (I_{ON}), and off-state current (I_{OFF}). Temperature-dependent variability is systematically analyzed to further elucidate the behavior of these advanced devices. To capture the dynamic effects of aging, the reaction-diffusion (RD) framework—which accounts for defect generation due to Negative Bias Temperature Instability (NBTI)—is implemented in TCAD, enabling detailed modeling of trap generation and the corresponding V_{TH} shifts in p-type transistors under varying gate stress biases (V_{GSTR}) and operating temperatures. At the circuit level, a full array of 6T-SRAM cells with the requisite peripheral circuits is simulated using SPICE after careful calibration of the the industry-standard compact model of gate-all-around (BSIM-CMG) against the TCAD data. Variability analysis reveals that the access disturb margin achieves a cell sigma (μ/σ) of 17.4 at nominal supply voltage, significantly exceeding the 6σ robustness criterion for read disturbances. Moreover, as the operating temperature increases from 300 K to 398 K, the read static noise margin (RSNM) and hold static noise margin (HSNM) degrade by 13.7% and 6.37%, respectively, while the write static noise margin (WSNM) improves by 18.3 %. These findings provide critical insights into the design trade-offs and reliability challenges of CFET-based SRAMs.

INDEX TERMS Aging, CFET, compact modeling, process variation, SRAM, TCAD

I. INTRODUCTION

MOORE’S law has driven the scaling of semiconductor technology for over five decades, facilitating continuous enhancements in the performance of integrated circuits (ICs) by preserving the current density per unit silicon area [1]. However, traditional scaling of transistors has encountered significant challenges, reaching a miniaturization limit beyond the 20 nm technology node. This limitation primarily arises from weak electrostatic control over the transistor channel, which results in increased leakage power. The scaling challenges associated with planar transistors have accelerated the shift towards three-dimensional (3-

D) architectures, such as FinFETs [2] and gate-all-around (GAA) nanosheet FETs (NSFETs) [3] at advanced technology nodes. These advanced structures provide enhanced electrostatic control through their multi-gate configurations. NSFETs, in particular, achieve high current density per unit silicon area by vertically stacking nanosheets, thereby overcoming the lateral spacing limitations inherent in classical FinFET designs.

However, cell-level scaling is constrained not only by transistor dimensions but also by the lateral spacing between n-type and p-type devices (nFET & pFET), which is restricted by lithographic limitations [4]. In high-density

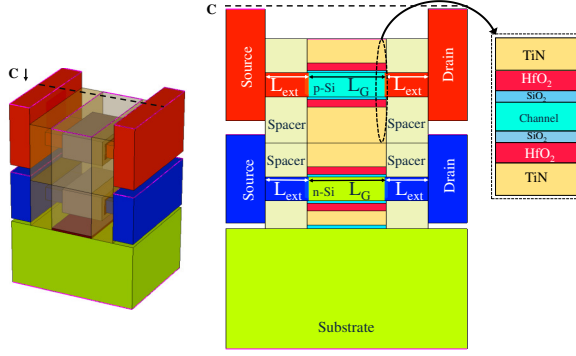


FIGURE 1. 3-D schematic of the vertically stacked CFET structure modeled in TCAD. The device dimensions are listed in Table. 1.

6T-static random access memory (SRAM) cells, progress in cell height reduction has lagged behind logic scaling due to routing congestion and pin accessibility challenges [5], [6]. To overcome this challenge, the complementary FET (CFET) architecture has emerged as a promising solution, enabling approximately 50% reduction in silicon footprint by placing the nFET and pFET on top of each other (Fig. 1), thereby eliminating the lateral spacing constraint in standard cell layouts [7]–[9].

Although CFET has attracted significant interest in the VLSI community for its potential performance enhancements, its variability and reliability challenges remain largely underexplored. As fabrication processes become more complex, CFETs are anticipated to exhibit greater design-time variability compared to FinFETs and NSFETs. As transistor dimensions continue to scale down at advanced technology nodes, the impact of process variations has become more pronounced, particularly in circuits that require high integration density. These design-time variations affect critical transistor characteristics, such as threshold voltage (V_{TH}), ON-state current (I_{ON}), and OFF-state current (I_{OFF}), which directly influence the performance of circuits. For instance, SRAM cells, often designed with the smallest possible transistor size to maximize integration density, are particularly susceptible to variability, as their noise margins are highly sensitive to fluctuations in the V_{TH} of the transistors. The primary sources of random variability in semiconductor devices include line edge roughness (LER) [10], [11], gate edge roughness (GER) [12], work-function variability (WFV) induced by intrinsic metal gate granularity (MGG) [13], [14], and random dopant fluctuations (RDF) [15]. Although RDF and LER are significant sources of variability in nanoscale devices, MGG-induced WFV is the primary contributor to V_{TH} fluctuations in undoped GAA NSFET [15]–[17]. Additionally, negative bias temperature instability (NBTI)-induced aging has long been a significant run-time reliability challenge for pFETs, spanning from silicon oxynitride (SiON)-based planar transistors [18] to high-k metal gate and replacement metal gate-based FinFETs [19], [20]. The 3-D stacking of transistors and the GAA architecture in CFETs are likely to exacerbate Joule heating effects, making them more susceptible to aging [21]. NBTI arises when a negative

TABLE 1. Physical Parameters of the Device as in [9]

Parameters	Value
Gate length	15 nm
Nanosheet width	16 nm
Nanosheet thickness	6 nm
SiO ₂ thickness	0.9 nm
Channel doping concentration	$1 \times 10^{15} \text{ cm}^{-3}$
Source/Drain doping concentration	$1 \times 10^{20} \text{ cm}^{-3}$
Nominal supply voltage	0.75 V

bias is applied as a stress to the gate of a pFET relative to its other terminals. This results in the accumulation of positive charges at the interfaces of the gate stack, leading to shifts in key transistor parameters such as V_{TH} , drain current (I_{DS}), sub-threshold swing (SS), and transconductance (g_m). The accumulation of positive charges and the resulting shift in transistor parameters depend on both the applied V_{GSTR} and the operating temperature. While parameter shifts recover partially once the stress is removed, AC stress causes less degradation than DC stress [22], [23]. Therefore, both the stress magnitude and its duration, along with the type of stress, are critical factors in the run-time degradation process.

In this work, we develop a simulation framework using Sentaurus TCAD to evaluate the impact of process variations and NBTI on CFETs. The framework incorporates the statistical impedance field method (sIFM) for variability analysis, which calculates the linear response of a reference device to various perturbations, such as geometric variations, WFV, and RDF [24]. We analyzed the impact of process variations for CFETs operating under four different temperatures (230 K, 300 K, 348 K, and 398 K). To evaluate the impact of design-time variations on circuits, we have developed a SPICE model for the underlying CFET using the industry-standard BSIM-CMG model [25]. Using the developed model, we have conducted Monte Carlo SPICE simulations for an SRAM array. The transistor-level variations in SPICE are modeled as fluctuations in V_{TH} , consistent with the variations observed in the TCAD simulations. For NBTI analysis, we utilize the reaction-diffusion (RD) model implemented in Sentaurus TCAD with the multi-state configuration (MSC) framework and the capture emission depassivation (CED) model for inversion hole and IL field-assisted breaking of interfacial Si-H bonds [26]. The RD model computes trap density (ΔN_{IT}) at the channel/interlayer (IL) and IL/high-k interfaces, while the Transient Trap Occupancy Model (TTOM) estimates trap charge occupancy and its contribution to ΔV_{TH} .

Our key contributions within this work are:

- We develop a 3-D TCAD model for CFETs, calibrated against experimental data [9].
- Using the calibrated TCAD deck, we conduct variability analysis with the sIFM model of TCAD to assess the impact of process variations on device performance. We study the MGG-induced WFV and RDF as primary

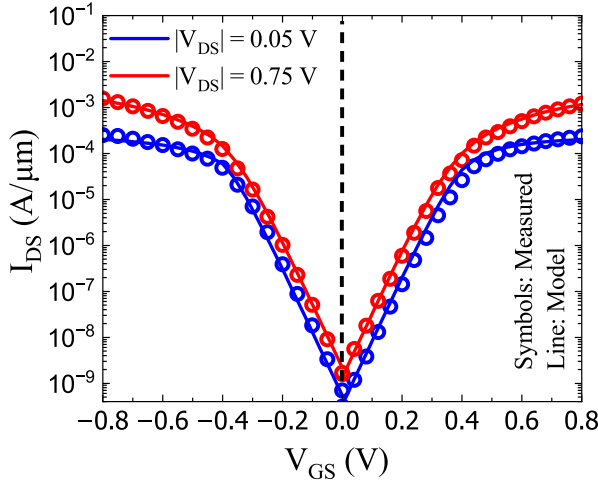


FIGURE 2. Calibration of the I-V characteristics with experimental data [9] for a vertically stacked CFET device with a gate length of 15 nm.

- variability sources under different operating temperatures: 230 K, 300 K, 348 K, and 398 K.
- We also employ the MSC framework and CED model in TCAD to evaluate the impact of NBTI on ΔV_{TH} at two different temperatures (398 K and 473 K) under various V_{GSTR} .
 - Finally, we calibrate the industry-standard BSIM-CMG model using TCAD-simulated characteristics across a range of temperatures. Using the calibrated model, we assess the effects of design-time variations on SRAM cell performance, with particular emphasis on noise margins and read/write delays. Additionally, we examine the impact of temperature fluctuations as a run-time variation affecting SRAM performance.

II. MODELING METHODOLOGY

A. TCAD Calibration

Fig. 1 presents the 3-D and cross-sectional views of the CFET device structure, with the relevant design parameters shown in Table. 1. This CFET model is built and simulated using the Sentaurus TCAD tool from Synopsys [27] to closely match the experimental data [9]. The Philips unified model is used to capture the mobility variations with temperature, and a high-field saturation model is used to capture velocity saturation at high drain biases. In addition, the Shockley-Read-Hall, Auger, and Band-to-Band recombination models are integrated to account for trap-induced tunneling currents. The simulation framework incorporates Fermi-Dirac statistics and a bandgap narrowing model to model the carrier densities accurately. Critical parameters, including source/drain doping profiles, source-drain series resistance, and gate metal work function, are carefully tuned to achieve consistency with experimental results, as illustrated in Fig. 2.

B. NBTI Modeling

NBTI degradation is primarily driven by the dissociation (stress) and passivation (recovery) of Si-H bonds at the gate/insulator interface, followed by hole trapping at the

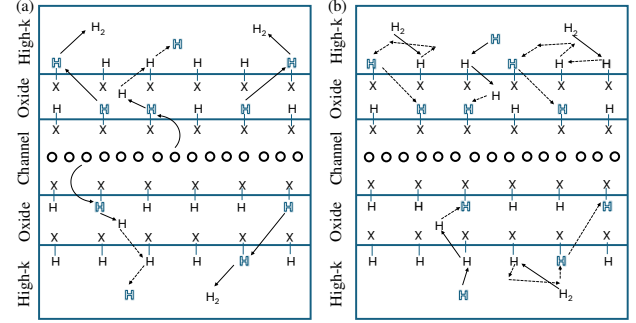


FIGURE 3. Representational view of the bond dissociation and passivation [28] during (a) stress, and (b) recovery.

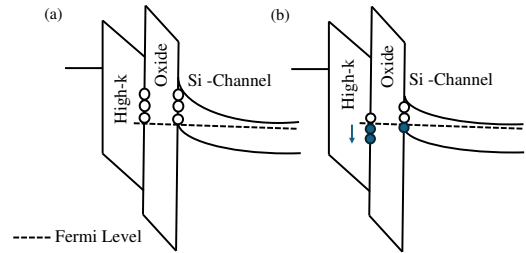


FIGURE 4. Representational energy band diagrams of the transistors during (a) stress and (b) recovery. During recovery, some traps move below the Fermi level, leading to the de-trapping of holes. This process contributes to the partial recovery of the ΔV_{TH} [28].

resulting dangling bonds (Fig. 3). Two key mechanisms govern this process: (1) the reaction and diffusion of hydrogen species within the interfacial layer, modeled using the RD approach within the MSC framework in TCAD, and (2) bond dissociation, captured through the CED model integrated into TCAD, as detailed in [29]. The bond dissociation rate exhibits an exponential dependence on both the applied stress field and temperature. The diffusion mechanism is strongly influenced by the stress duration, with longer stress periods allowing increased time for diffusion and subsequent reactions. We utilize the key parameters of the CED model like the prefactor (KF10), temperature activation energy (EAKF1), field acceleration constant (Γ_0), and bond polarization factor (α) [29], [30] to characterize the breaking of Si-H bonds. The traps formed due to bond dissociation are considered donor-like (positively charged when unoccupied). During stress, only those traps situated energetically above the substrate's Fermi level become positively charged and contribute to NBTI, as illustrated in Fig. 4 (a). When the gate bias magnitude decreases during recovery, some of these traps fall below the Fermi level, as shown in Fig. 4 (b). These traps capture electrons, become neutral, and no longer contribute to NBTI, even though they have not yet been passivated through back-diffusion of hydrogen, as described by the RD model. This behavior is accounted for using the TTOM in TCAD, which includes the Activated Barrier Double Well Thermionic (ABDWT) model [29], [30].

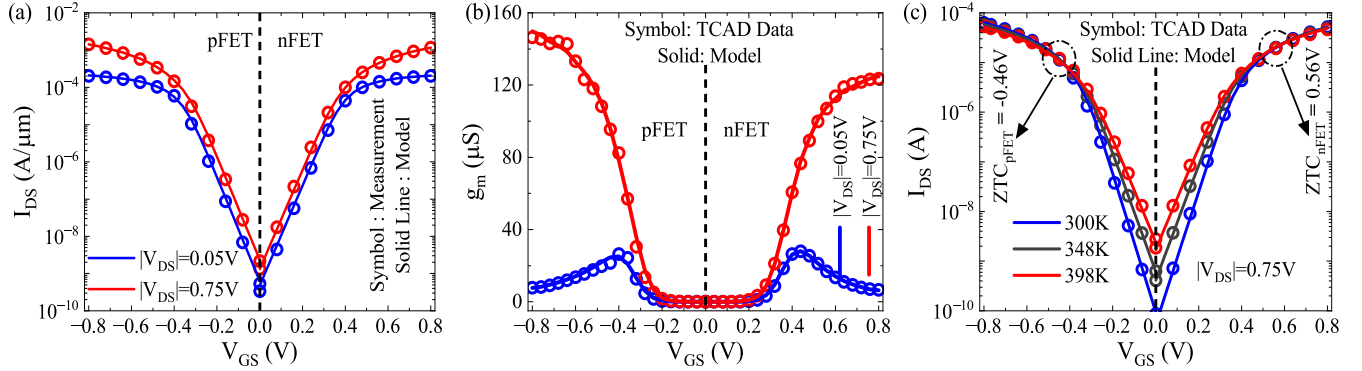


FIGURE 5. The calibrated BSIM-CMG model demonstrates excellent agreement with the TCAD simulated data: (a) transfer characteristics at 300 K, (b) transconductance (g_m) versus V_{GS} at 300 K, and (c) transfer characteristics at three different temperatures. The zero transconductance (ZTC) point for the n-FET is observed at 0.56 V, while for the p-FET, it is at -0.46 V.

C. Variability Modeling

IFM in TCAD offers an efficient and accurate approach for the analysis of statistical variability [24]. It models randomness as a perturbation from a reference device, avoiding the need to repeatedly solve the full nonlinear Poisson and drift-diffusion equations for multiple stochastic instances. Instead, a single 3-D TCAD solution is obtained for the reference device, after which current fluctuations due to perturbations are calculated using linear response theory and a Green's function approach [31], [32]. A key benefit of IFM is its scalability to complex 3-D geometries, making it suitable for advanced device structures such as FinFETs and CFETs. Sentaurus TCAD provides three IFM variants for statistical variability analysis: noise-like, statistical, and deterministic IFM [27]. This work focuses on applying the statistical IFM to the 3-D CFET device. For example, if we consider the RDF case, once the solution for current is obtained for the reference device with doping profiles $N_{ref}^S(\vec{r})$, the method calculates the current deviation at contact 'c' as a linear response to the perturbations [27], which is represented as:

$$\delta I_{\nu,c} = \sum_s \int G_c^S(\vec{r}) \delta N_{\nu}^S(\vec{r}) d^3r \quad (1)$$

$$\delta N_{\nu}^S(\vec{r}) = N_{\nu}^S(\vec{r}) - N_{ref}^S(\vec{r}) \quad (2)$$

where $N_{\nu}^S(\vec{r})$ is one particular doping profile randomization with the sample index ' ν ' for the doping species ' s '. The Green's function $G_c^S(\vec{r})$, also known as the impedance field, is independent of doping profile randomizations. It is calculated from the TCAD solution of the reference device and needs to be computed only once, regardless of the number of doping randomizations considered. $N_{\nu}^S(\vec{r})$ in (1) is used for statistical sampling across multiple doping randomizations. This approach assumes spatially uncorrelated doping, where the number of dopants in a given volume follows a Poisson distribution based on the average dopant concentration. Using this assumption, Sentaurus Device generates random acceptor and donor atom counts at each vertex by sampling within the corresponding Voronoi volume, then converts these counts to concentrations by normalizing with the

Voronoi volume [27]. For MGG-induced WFV, sIFM models variability by randomly distributing metal grains along the relevant metal contact. The user specifies the average grain size (5 nm in this work), the number of grain types, and the associated work function and probability for each type [27].

D. SPICE Compact Modeling

To assess CFET performance and analyze the impact of process variations on circuit variability and reliability, it is essential to develop SPICE compact models for the underlying transistors. We use the BSIM-CMG compact model [25], widely recognized as the industry standard for GAA transistor technology, to simulate the underlying transistors of our CFET and calibrate against the measurement data [9]. Our model extraction procedure begins with defining key process parameters, including equivalent oxide thickness (EOT), channel doping concentration (NSUB), channel length (L), nanosheet width (WGAA), and nanosheet thickness (TGAA). The CFET devices in this study have an L of 15 nm, WGAA of 16 nm, and TGAA of 6 nm. After assigning the structural parameters to the model, we extract critical parameters that impact sub-threshold and linear behavior. Specifically, sub-threshold performance is fine-tuned using model parameters such as PHIG (work function), CIT (interface traps), and CDSC (source/drain coupling capacitance). To model mobility degradation, we calibrate mobility-related parameters, including U0 (low-field mobility), UA and EU (phonon and surface roughness scattering), UD (Coulomb scattering), and ETAMOB (effective field parameter) under moderate inversion conditions at low drain biases. At the strong inversion, we extract source/drain series resistances (RSW/RDW) to accurately characterize linear operation. At high drain bias ($V_{DS} = 0.75$ V), sub-threshold swing (SS) and drain-induced barrier lowering (DIBL) are modeled by adjusting the CDSCD parameter to reflect drain bias sensitivity affecting SS and tuning ETA0 for accurate DIBL representation. High-field induced velocity saturation effects are captured by optimizing the VSAT parameter. Additionally, channel length modulation is accounted for by adjusting the PCLM parameter at high drain bias. The

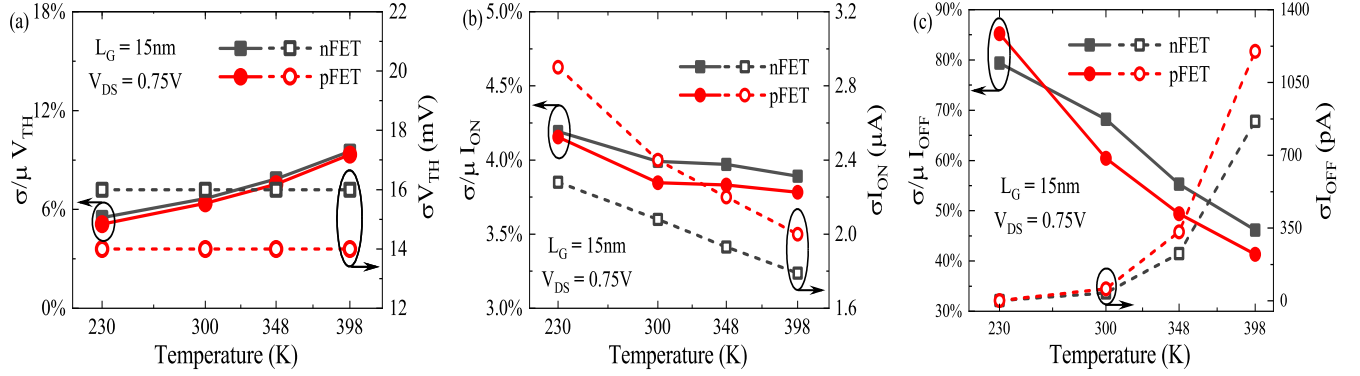


FIGURE 6. WVF and RDF-induced variations in (a) V_{TH} , (b) I_{ON} , and (c) I_{OFF} at different temperatures for both nFET and pFET. V_{TH} and I_{OFF} show higher variation compared to I_{ON} . As V_{TH} decreases with temperature, its sensitivity to variations increases, leading to a higher σ/μ at higher temperatures. In contrast, higher temperatures significantly increase I_{OFF} , reducing its sensitivity to variations.

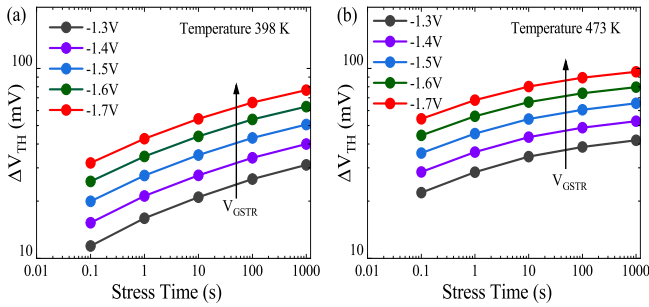


FIGURE 7. ΔV_{TH} over time under stress at different V_{GSTR} (a) for $T = 398$ K, (b) for $T = 473$ K.

model's accuracy compared to measured characteristics is demonstrated in Fig. 5 (a) & (b). To accurately model the impact of temperature on the characteristics of the transistors, temperature-dependent parameters are extracted and calibrated. Specifically, KT1 and TSS are adjusted to capture the temperature-induced shifts in subthreshold swing (SS) and V_{TH} . For mobility degradation in the linear operating regime, temperature coefficients such as UCSTE, UTL, EMOBT, EU1, and UA1 are optimized to reflect the temperature dependence of carrier mobility. At saturation ($V_{DS} = 0.75$ V), the temperature dependence of the carrier velocity saturation is captured using the parameters AT, A11, and A21. Fig. 5 (c) illustrates that the developed compact model shows excellent agreement with the TCAD data for three temperatures: 300 K, 348 K, and 398 K. coefficientsted zero-temperature coefficient (ZTC) of the nFET and pFET are 0.56 V and -0.46 V, respectively.

III. RESULTS AND DISCUSSIONS

A. Device-level Variability

This work investigates the impact of WVF and RDF on the characteristics of CFETs at $V_{DD} = 0.75$ V using the sIFM within a TCAD framework, as described in Section. II (C). For WVF analysis, the gate contact is defined as the surface on which grain size and corresponding work function variations are imposed. The gate work function values are varied from 4.42 eV to 4.62 eV. For RDF analysis, a Gaussian distribution is used to model doping variations for both acceptor (nFET) and donor (pFET) concentrations.

The randomization is conducted over 10,000 samples to ensure statistical significance. The combined impact of these variations on V_{TH} , I_{ON} , and I_{OFF} , is presented in Fig. 6, which shows the standard deviation (σ) and normalized variability data (σ/μ) across multiple temperatures. The overall variation for nFET (pFET) devices results in standard deviations of 16 mV (15 mV) for V_{TH} , 2.1 μ A (2.8 μ A) for I_{ON} , and 35 pA (51 pA) for I_{OFF} at room temperature. The V_{TH} and I_{OFF} exhibit more variations compared to I_{ON} . Furthermore, as the V_{TH} of transistors decreases with increasing temperature, the sensitivity of V_{TH} to variations becomes more significant, as is evident from the larger σ/μ at higher temperatures (Fig.6 (a)). In contrast, at high temperatures, the leakage current increases substantially, thereby reducing the sensitivity of I_{OFF} to variations, is observed in Fig.6 (c).

B. Device-level BTI

Fig. 7 presents the TCAD-predicted V_{TH} degradation over time at temperatures of 398 K and 473 K, under different V_{GSTR} , utilizing the TTOM-enabled RD model. Fig. 8 (a) and (b) illustrate the V_{TH} shift as a function of V_{GSTR} and $1/k_B T$, respectively. The slope of the ΔV_{TH} versus V_{GSTR} yields the voltage acceleration factor (VAF) = 2.26, while the slope of V_{TH} versus $1/k_B T$ determines the activation energy (E_A) = 0.114 eV. These parameters (VAF and E_A) quantify the exponential dependence of degradation on V_{GSTR} and temperature. The degradation is governed not by the concentration of the generated interface traps but by the occupancy state of these traps. When the stress is removed, some of the generated traps fall below the Fermi level, rendering them inactive and no longer contributing to the degradation. Consequently, the device undergoes partial recovery once the applied stress is withdrawn. The TCAD-based NBTI modeling framework effectively captures this recovery behavior using the TTOM model, which accounts for bias-dependent trap occupancy, as illustrated in Fig. 9.

C. Circuit-level Variability

To evaluate the impact of variability on circuit performance, the BSIM-calibrated CFET model is employed to simulate a

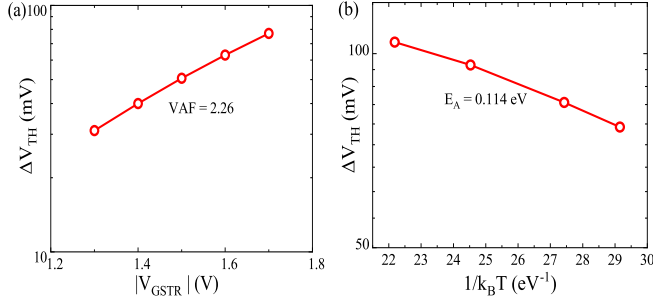


FIGURE 8. (a) V_{TH} over (V_{GSTR}) at a temperature of 473 K. The VAF extracted from the slope of the plot is 2.26, (b) V_{TH} vs. $(1/k_B T)$ at $V_{GSTR} = 1.7$ V. The activation energy (E_A) extracted from the plot is 0.114 eV.

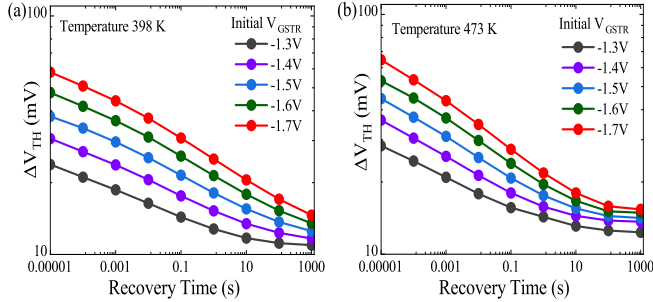


FIGURE 9. Recovery of ΔV_{TH} over time after the removal of stress at (a) $T = 398$ K, (b) $T = 473$ K.

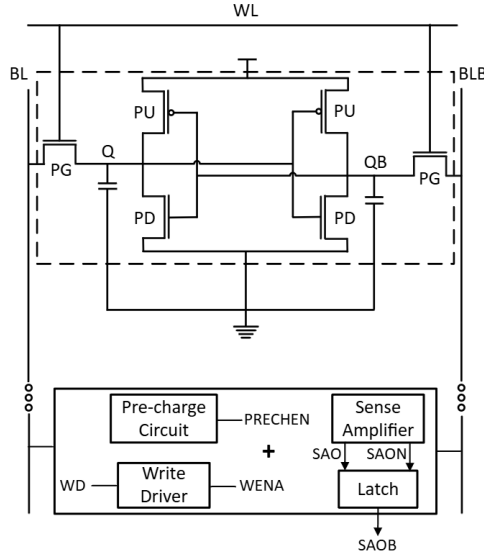


FIGURE 10. Schematic of a 6T-SRAM cell with its peripheral circuitry. The pull-up (PU), pull-down (PD), and pass-gate (PG) transistors facilitate data storage and access. The bitline (BL) and complementary bitline (BLB) connect with the precharge circuit, write driver, and sense amplifier. The wordline (WL) controls the PG transistors, enabling communication between the BL/BLB and the cross-coupled inverters that retain the stored data.

6T-SRAM framework. The simulation framework consists of an array of 32 SRAM cells connected to peripheral circuits responsible for read and write operations. The peripheral circuitry includes a pre-charge unit that initializes the BL/BLB for reading, a sense amplifier and latch that detect and hold voltage changes on BL/BLB during the read cycle, and a write driver with sufficient strength to overwrite the cell. A schematic overview of the SRAM framework is shown in

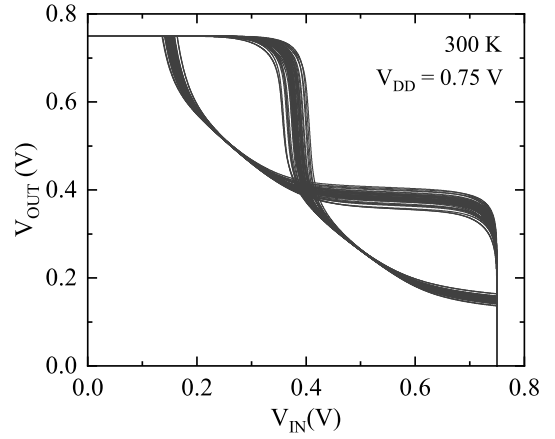


FIGURE 11. Read operation butterfly curve variations of the 6T SRAM cell due to process-induced fluctuations in the underlying transistors. The curves are generated through SPICE Monte Carlo simulations with 1000 iterations, employing the calibrated BSIM-CMG model and incorporating V_{TH} variations derived from TCAD simulations.

Fig. 10. The impact of variability is analyzed by performing Monte Carlo SPICE simulations for 1000 samples. Design-time variability is introduced by fluctuating the V_{TH} of the transistors within the SPICE model, with a standard deviation of 16 mV for nFETs and 15 mV for pFETs (as obtained from the TCAD simulations).

The robustness of SRAM cells against noise and voltage fluctuations is measured using the static noise margin (SNM), which we calculate from the butterfly curve [33]. Variations in the V_{TH} of the transistors affect the SNMs of the SRAM cell. Fig. 11 illustrates the variations in the read butterfly curve for the SRAM cell, obtained through Monte Carlo simulations with 1000 iterations at 300 K. These process-induced fluctuations result in standard deviations of approximately 8.3 mV, 15 mV, and 4.2 mV for the read, write, and hold SNMs, respectively, as shown in Fig. 12. Fig. 13 presents the distributions of read and write delays at 300 K. The results indicate that the standard deviations for the read and write delays are 1.46 ps and 1.23 ps, respectively. The stability of the read and write margins are evaluated using the cell sigma (μ/σ) of the RSNM and WSNM, which are 17.4 and 11.2, respectively. These values are well above the 6σ target required for ensuring robust access disturb margin [34]. Table 2 summarizes the effect of temperature as the run-time variability on the delay and SNMs of the SRAM cell. As the temperature increases, the operational delay of the SRAM slightly rises due to a reduction in the I_{ON} of the transistors. Additionally, the RSNM and HSNM decrease by an average of 13.7% and 6.37%, respectively, as temperature increases from 300 K to 398 K. In contrast, WSNM increases by an average of 18.3% with rising temperature, primarily due to the greater degradation of pFETs compared to nFETs (access transistors).

IV. CONCLUSION

This work investigates the impact of WFV and RDF as process variation sources on GAA-based CFET transistors using

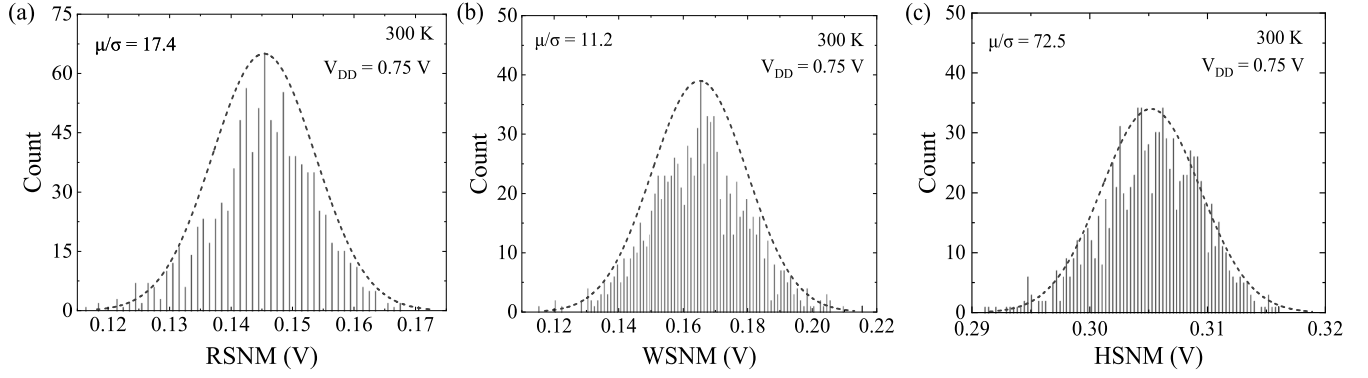


FIGURE 12. Distribution of static noise margins for 6T SRAM at $V_{DD} = 0.75$ V and $T = 300$ K: (a) Read Static Noise Margin (RSNM), (b) Write Static Noise Margin (WSNM), and (c) Hold Static Noise Margin (HSNM).

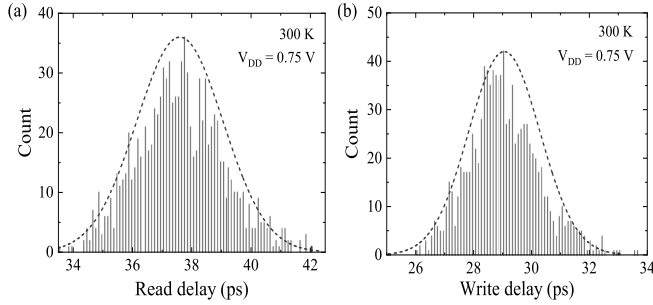


FIGURE 13. Distribution of delays of 6T SRAM operation induced by process variations in the underlying transistors: (a) read delay, and (b) write delay.

TABLE 2. Impact of Temperature on CFET-based 6T-SRAM's performance.

	T=300 K	T=348 K	T=398 K
Read Delay (ps)	37.59	37.88	39.11
Write Delay (ps)	34.47	34.44	34.94
HSNM (V)	0.306	0.293	0.280
RSNM (V)	0.146	0.134	0.121
WSNM (V)	0.164	0.183	0.205

3-D TCAD simulations. The extracted average (considering both nFET and pFET) standard deviations of ΔV_{TH} , ΔI_{ON} , and ΔI_{OFF} are 15.5 mV, 2.45 μ A, and 43 pA, respectively. The TCAD-based NBTI degradation analysis under stress conditions at 398 K and 473 K yields VAF and E_A values of 2.26 and 0.114 eV, respectively. Variability analysis of SRAM simulations reveals that design-time process variations lead to standard deviations of 8.3 mV, 15 mV, and 4.2 mV for read, write, and hold SNMs, respectively, at nominal supply voltage and room temperature. The SRAM achieves a cell sigma of 17.4 and 11.2 for read and write operations at nominal supply voltage, respectively, exceeding the 6σ target for robust disturb margin. Additionally, run-time variability due to temperature fluctuations indicates that as the temperature rises from 300 K to 398 K, the RSNM and HSNM decrease by an average of 13.7% and 6.37%, respectively, while the WSNM improves by 18.3%.

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